



Hate Prototypes: Interpretable and Transferable Representations for Implicit and Explicit Hate Speech Detection

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Introduction

- Online hate speech has significant societal impact, including psychological harm and disruption of online communities
- Hate speech detection models achieve **strong in-domain performance** but exhibit **limited generalization to implicit**, multilingual, and out-of-domain settings
- Hate speech detection models remain **computationally expensive**, limiting their applicability in real-time moderation scenarios

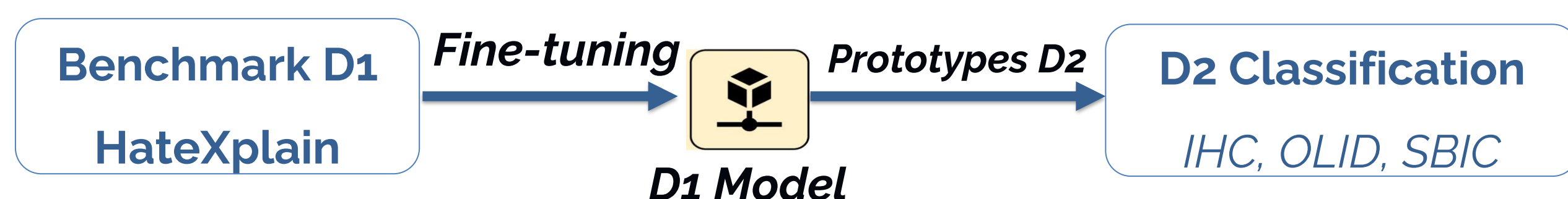
Contributions

- We analyze the role of HatePrototypes in the **transferability** of LMs optimized for implicit hate detection, explicit hate detection, and safety moderation
- We show that HatePrototypes are transferable between **implicit and explicit hate messages**
- We further explore **early exiting** in models fine-tuned on implicit and explicit hate speech tasks, demonstrating how layer wise prototype construction can enhance efficiency and performance

Experimental Setup

- Models:** OPT-125M, BERT-125M, LLAMA-GUARD-1B, BLOOMZ-3B-GUARD
- Benchmarks:**
 - IMPLICIT HATE CORPUS (implicit)
 - SOCIAL BIAS INFERENCE CORPUS (implicit)
 - HATEXPLAIN (explicit)
 - Offensive Language Identification Dataset (explicit)

Cross-domain prototype transfer



- Prototype-based classification significantly improves **cross-domain performance** for both BERT and OPT across all settings
- Prototype-based performance approaches **fine-tuned in-domain baselines** without statistically significant differences in most cases
- Performance gains are **stronger in cross-domain settings** than in-domain, indicating improved transferability
- Prototypes derived from implicit datasets (e.g., IHC) remain effective for **explicit hate classification**, demonstrating cross-type transfer

FT-Eval.	F1	Δ F1
HX-HX	77.12	-0.83
HX-OLID	67.39*	+20.42
HX-IHC	61.48*	+3.32
HX-SBIC	71.38*	+28.02
IHC-HX	62.51*	+3.71
IHC-OLID	60.77*	+9.75
IHC-IHC	73.78	+0.26
IHC-SBIC	68.61	+4.04
SBIC-HX	67.31*	+16.75
SBIC-OLID	72.78*	+3.32
SBIC-IHC	60.66*	+6.53
SBIC-SBIC	82.37*	+0.20

Prototype Transfer

- Prototypes built from other datasets **achieve near in-domain performance** → up to 95–100% of fine-tuned performance retained
- Implicit** datasets (IHC) **transfer best** across domains, even to explicit hate tasks

	BERT					OPT			
	HX	IHC	SBIC	OLID		HX	IHC	SBIC	OLID
Encoder/Eval domain									
OLID	100.0	97.3	79.6	90.1		100.0	92.9	58.4	72.4
IHC	99.6	100.0	96.7	98.6		94.2	100.0	64.0	82.9
SBIC	81.4	94.5	100.0	96.4		83.4	98.9	100.0	96.9
OLID	95.0	99.4	100.0	100.0		96.3	95.6	94.7	100.0
	Prototype domain					Prototype domain			

Relative cross-dataset transfer F1-scores. Values show how much of each model's in-domain performance carries over to other datasets.

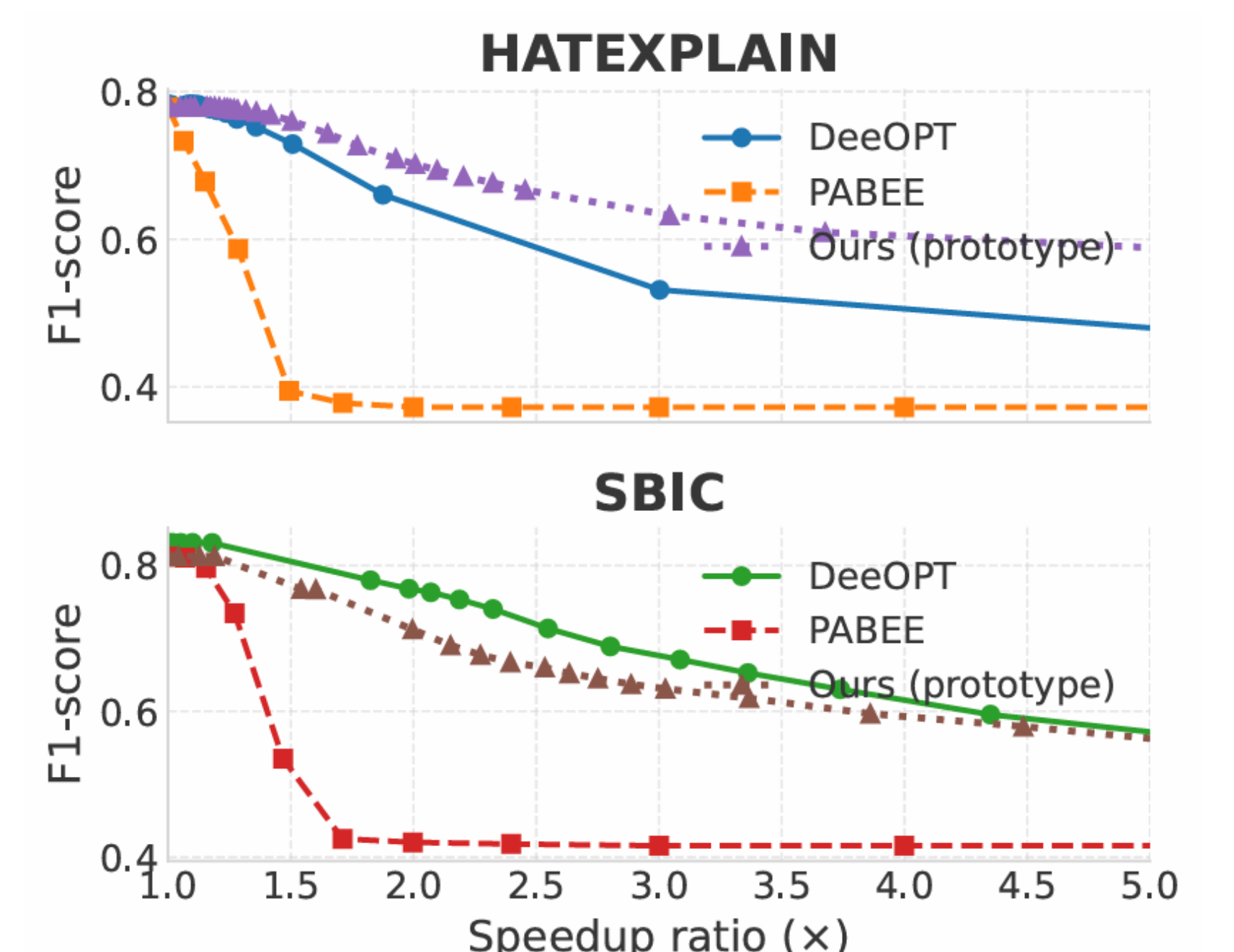
Prototype Classification for Guard Models

- Prototypes **improve model performance**, with large gains (+18 F1 on SBIC with LLaMA-Guard, +11 F1 on IHC with BLOOMz-Guard)
- Effective across both **implicit and explicit** hate, enabling out-of-domain transfer without fine-tuning

FT-Eval.	Acc. Base	Acc. P.	F1 Base	F1 P.
LLaMA-1B-Guard				
HX	66.74	67.95	64.03	67.59
OLID	46.86	71.29	46.36	60.96
IHC	52.76	64.28	52.66	62.27
SBIC	52.31	74.07	52.14	70.33
BLOOMz-3B-Guard				
HX	59.30	61.75	59.29	61.43
OLID	80.35	77.49	68.53	73.50
IHC	65.64	63.36	49.49	60.92
SBIC	44.43	55.01	44.16	54.45

Early-Exiting with Prototypes

- Prototype-based early exiting achieves up to **~1.5× speed-up** with no significant drop in F1-score, outperforming patience-based methods



Conclusion and Future Work

We introduce HatePrototypes, a parameter-free approach for implicit and explicit hate speech classification based on class-level representations. We show that prototypes improve out-of-domain performance without degrading in-domain accuracy and support efficient inference via early exiting. We release the framework and prototype resources to enable further analysis of hate representations across models and datasets.